

Chapter1 Data Storage

Bits and Their Storage

- **Bits:** Binary Digit(0 or 1)
- **Bit pattern:** Represent some information. e.g. Numbers, Text characters, Images, Sound etc

Boolean Operation

- **Operations that manipulate true/false values.**
- **Specific Operations:** AND, OR, XOR, NOT

	(1,1)	(1,0)	(0,1)	(0,0)
AND	1	0	0	0
OR	1	1	1	0
XOR	0	1	1	0

Gates and Flip-Flops

- **Gates:** A device that computes a Boolean operation.
 - Usually constructed out of small electronic switching circuits called transistors.
 - Provide the building blocks from which computers are constructed.
 - **VLSI** (Very Large Scale Integration)

AND



Inputs	Output
0 0	0
0 1	0
1 0	0
1 1	1

OR



Inputs	Output
0 0	0
0 1	1
1 0	1
1 1	1

NAND



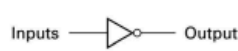
Inputs	Output
0 0	1
0 1	1
1 0	1
1 1	0

XOR



Inputs	Output
0 0	0
0 1	1
1 0	1
1 1	0

NOT



Inputs	Output
0	1
1	0

NAND

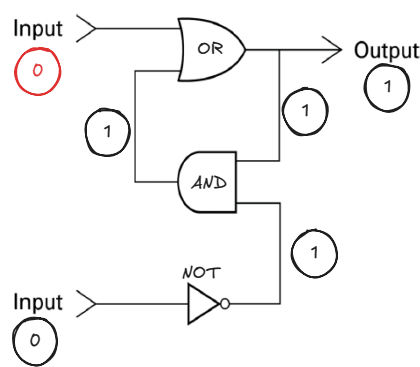
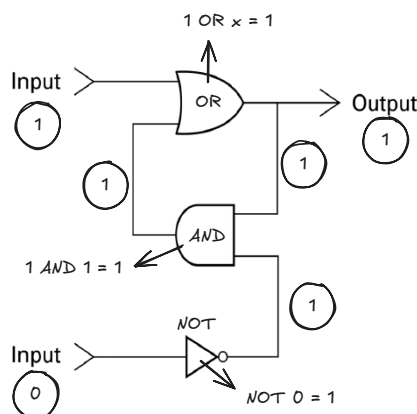


Inputs	Output
0 0	1
0 1	0
1 0	0
1 1	0

- **Flip-flop:** A circuit built from gates that can store one bit.
 - One input line is used to set its stored value to 1.
 - One input line is used to set its stored value to 0.
 - While both input lines are 0, the **most recently stored value is preserved**

First, the upper input is 1.

When upper input become 0, the output is still 1.

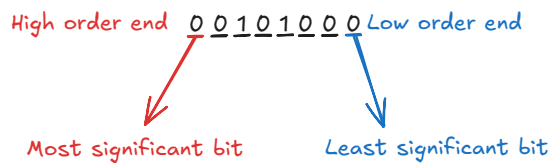


Hexadecimal Notation

- **A shorthand notation for long bit patterns**
 - Divides a pattern into **groups of four bits** each
 - Represents each group by a single symbol. e.g. 11010111 -> D7

Main Memory Cells

- **Cell:** A unit of main memory. (Typically 8bits which is one byte)
 - Most Significant Bit: The most left bit. ^[1]
 - Least Significant Bit: The most right bit.



- **Address:** To identify individual cells in a computer's main memory
 - It consists of numbers.
 - These numbers are assigned consecutively starting at zero.
 - This numerical ordering not only identifies each cell but also establishes an **order**.

Memory Terminology

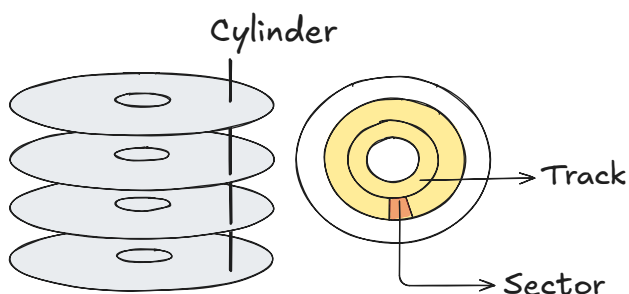
- **Random Access Memory (RAM):** Main memory can access any cells as required.
- **Dynamic Memory (DRAM):** RAM composed of **volatile memory** (揮發型記憶體). ^[2]

Measuring Memory Capacity

- **Kilobyte:** 2^{10} bytes
- **Megabyte:** 2^{20} bytes
- **Gigabyte:** 2^{30} bytes

Mass Storage

- **Advantages over main memory**
 - Less Volatility
 - Large storage capacities
 - Low cost
 - In many cases can be removed. e.g. Magnetic disks, Magnetic tapes, disks, CDs, DVDs, Flash drives, Solid-state disks
- **Magnetic Disks**
 - **Track:** Data is recorded on a disk's surface along concentric circles.
 - **Sector:** Track is divided into small arcs.
 - **Cylinder:** Multiple platters (disks), the collection of all tracks that are accessible at a single radial position of the read/write heads.
 - **Zoned-Bit Recording:** Several adjacent tracks are collectively known as zones. → Improve usage efficiency.



- **Flash Drives**
 - **Flash Memory:** Circuits that trap electrons in tiny silicon dioxide chambers. → Higher Speed
 - Repeated erasing slowly damages the media → **Not suitable for general main memory applications.**
 - Application: Flash Drives, SSD, Secure Digit

Representing

- **Text:** Each character is assigned a unique bit pattern.

7 bits ASCII English



8 bits ISO-Extended ASCII English + Western European



21 bits Unicode(UTF-8) ALL Language

	ASCII	ISO's ASCII	Unicode(UTF-8)
Bit Length	7 or 8	8	21
Number of Symbols	128 or 256	256	>143000
Language Support	English	Western European	All Language
Compatibility	Basic	Compatible with ASCII	Compatible with ASCII
Multi-language Support	X	X	O

• Numeric Value

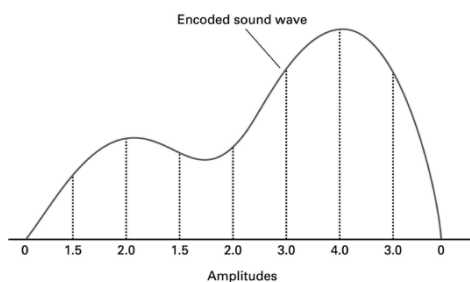
- Binary Notation: Uses bits to represent a number in base two.
- Limitation:
 - Overflow: Occurs when a value is too big to be represented.
 - Truncaion: Occurs when a value cannot be represented accurately. e.g. Exceeded storage space or display limit.

• Image

- Bit Map: Consists of **pixels**.
 - **If rescale the image, it will appear grainy.**
 - Binary Image: 1 bit (1 is black, 0 is white)
 - Grayscale Image: 8 bits (Each pixel can represent 256 shades of gray)
 - **RGB: 3 bytes** (Each pixiel consists of Red, Green, Blue, each using 1 byte)
 - Luminance and chrominance: Each pixel consists of brightness^[3], blue chrominance(Cb), red chrominance(Cr).^[4]
- Vector: Use **analytic geometry**^[5] to endcode.
 - **The image doesn't appear grainy when it is rescaled.**
 - TrueType: Describe text symbol which developed by Apple and Microsoft.
 - PostScript: Describe text symbol which developed by Adobe.
 - CAD syetem: Draw and manipulate **2D and 3D models**.

• Sound

- Sample: The amplitude of the sound wave is sampled and recorded regularly.
 - Used for high quality recording
 - Records actual audio(實時錄製)



- MIDI: **Encodes** which instrument plays which note and for how long.
 - Used in the music synthesizers. e.g. Electric keyboard, video game audio
 - MIDI files much smaller than sampled audio files.

Store Integers

- **Two Complement Notation:** The most popular means of representing integers value.

Easy Sign Detection

most left bit is represent + (0), - (1)

7: 0111
 ↓ reverse each bit
 1000
 ↓ +1
 -7: 1001

Natural Overflow Behavior

4 bits: _ _ _ _ (-8~7)
 max integer: 0111(7)
 ↓ +1
 min integer: 1000(-8)

Excess Notation

- find the first pattern with 1 and represent it is zero.

Excess-4 notation(bias = 4)

Bit pattern	Value represented
111	3
110	2
101	1
100	0
011	-1
010	-2
001	-3
000	-4

first pattern with 1 → 100 = 4
 4 - 0 = 4
 bias = 4

→ -2 + bias(4) = 2, binary of 2 = 010
 so -2 in excess-4 is 010
 most left bit is represent +(1), -(0)

Natural Overflow Behavior:
 111(3) + 1 = 000(4)

Floating-Point Notation

Use 8 bits to represent FRACTION.

- Structure
 - Sign bit: 0(+) or 1(-)
 - Exponent field(3 bits)
 - Mantissa field(4 bits)
- Condition
 - Normalize form
 - Truncation error

Decode 0 1 1 0 1 0 1 1

Step1. Extract the mantissa and place a radix point.

.1011

Step2. Extract the exponent and use excess-4 to interpret it.

110 = 6, 6 - bias(4) = 2

10.11

↪ move the radix to the right by 2 bits

10.11 = $2\frac{3}{4}$

Encode $3\frac{5}{8}=11.101$

Step1. fill 11101 into mantissa field

1 1 1 0 1 —lost bit—→ Normalized Form: fill in the field starting with the leftmost 1

Step2. fill exponent field

11.10

result in Truncation Error

radix needs to move to the right by 2 bits

$$2 + \text{bias}(4) = 6 = 110$$

0 1 1 0 1 1 1 0

decode 01101110 = 3.5
original value = 3.625

-
1. If the contents of the cell were interpreted as representing a numeric value, this bit would be the most significant digit in the number.(權重最高) ↩
 2. Store bits as **tiny electric charges** that dissipate quickly. ↩
 3. The sum of the red, green, and blue components. ↩
 4. Determined by computing the difference between the pixel's luminance and the amount of blue or red light. ↩
 5. The image as a collection of geometric structures, such as lines and curves ↩